

How an LLM Works

If you've used AI, you've probably used an app like ChatGPT, Claude, or Gemini. The app is what you use. Under the hood, a **Large Language Model**, or **LLM** for short, does the work.

What's an LLM?



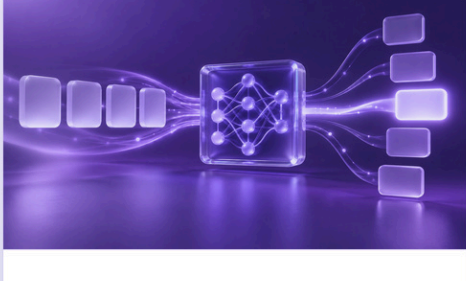
Large

Trained on huge amounts of text and code.



Language

Reads, writes, summarizes, translates, and explains.



Model

Predicts likely output from learned patterns.

✓ ChatGPT is the app. The LLM is the engine.

So how does an LLM actually turn your words into an answer?

When ChatGPT or Claude write a sentence, they're running math to predict the likely next words. They aren't looking up what your words mean; they're working out which words tend to follow which.

How do they do it? In two phases. First the model learns, once, by soaking up patterns from mountains of text. Then, every time you chat, it uses those patterns to build your answer one word at a time.

Learn Once. Answer Every Word.

LEARN ONCE

01 Training

The model learns from enormous amounts of data once, before you use it.

02 Patterns

Training turns examples into learned numerical patterns.

Patterns power every answer

ANSWER - EVERY WORD

03 Probability

For every next word, the model scores what is most likely.

04 Prediction

It chooses one likely next word, then runs the process again.

✓ Learn once. Use the patterns for every answer.

Four ideas carry it. Below, we follow one example, peanut butter, through all four.

01 Training

The model **teaches itself**: guess the next word, check, and nudge its numbers toward the right word.

How Training Works



1

Read

The model reads a training example with the answer included.



2

Guess

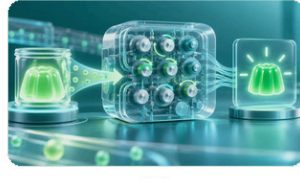
Given 'Peanut butter and ___,' the model guesses cloud.



3

Check

The example says jelly. Compare the guess with that word.



4

Adjust

Adjust the model's internal numbers to make jelly more likely in this situation.

NEXT EXAMPLE

✓ Repeat with more examples. The patterns build.

02 Patterns

So what is it actually learning? **Patterns**. Here's one you picked up as a child. Which word comes next?

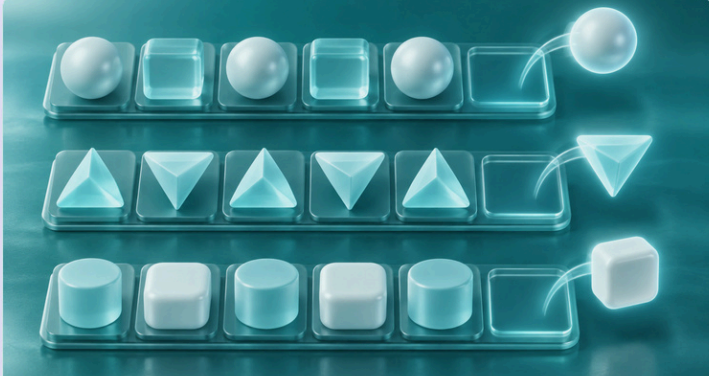
How AI Learns Patterns



One Familiar Pattern

Peanut butter and ___ → **jelly**

You knew it. So does AI.



Patterns Are Everywhere

Twinkle, twinkle, little ___ → **star**

Once upon a ___ → **time**

Better late than ___ → **never**

✓ AI learns patterns by working through billions of examples.

These are easy patterns you already know. AI also learns patterns in places you might not expect: how people explain ideas, ask questions, solve problems, and even misspell words.

03 Probability

AI doesn't make one guess. It scores **every** possible next word: a ranked list with a probability on each, and those numbers shift with the surrounding text.

Same Word. Different Odds.

I'd like to buy peanut butter and _____.

jelly		41%
bread		27%
bananas		16%
honey		5%

I'd like to buy a peanut butter and banana _____.

sandwich		54%
smoothie		16%
toast		9%
jelly		2%

✓ The surrounding words change the odds.

In this example, adding "banana" drops the probability of "jelly" from 41% to 2%.

04 Prediction

Probability handled one word. But your answer is hundreds of words long, so the model just repeats the move. Your phone does this when you write a text: it suggests a word, you tap it, it suggests the next.

One Word at a Time

I want to buy peanut butter and

→ **jelly**



I want to buy peanut butter and

jelly

→ **for**



I want to buy peanut butter and

jelly for

→ **lunch**

✓ Add a word. Use the updated sentence. Predict again.

A full paragraph runs this loop many times, fast enough to look like thought.

Training builds the patterns.

The model uses those patterns to build your answer, one word at a time.